

Problem Statement: Student Grading System

Scenario:

You are developing an automated grading module for a University Student Management System. The system needs to process a student's numerical exam mark and output their corresponding letter grade according to the official grading scale.

Grading Scale Criteria:

- **Invalid mark:** If the mark is greater than 100 or less than 0.
- **A+:** Mark is 80 or above (up to 100).
- **A:** Mark is 70 or above (up to 79).
- **A-:** Mark is 60 or above (up to 69).
- **B:** Mark is 50 or above (up to 59).
- **C:** Mark is 40 or above (up to 49).
- **D:** Mark is 33 or above (up to 39).
- **Fail:** Mark is below 33 (and non-negative).

Task:

Write a **C++ program** that takes an integer or floating-point input representing a student's mark and outputs the appropriate grade or error message.

Input Format:

- A single line containing an integer or floating-point number representing mark.

Output Format:

- Print the corresponding letter grade (A+, A, A-, B, C, D, Fail) or Invalid mark.

Sample Inputs & Outputs:

- **Sample 1:**
 - **Input:** 85
 - **Output:** A+
- **Sample 2:**
 - **Input:** 32
 - **Output:** Fail
- **Sample 3:**
 - **Input:** 105
 - **Output:** Invalid mark
- **Sample 4:**
 - **Input:** -5
 - **Output:** Invalid mark